Your grade: 100%

Your latest: **100%** • Your highest: **100%** • To pass you need at least 80%. We keep your highest score.

Next item

1. In a scenario where a classifier is trained using a generative model with incorrect assumptions about the underlying class structure, what is the likely impact on the classifier's accuracy when unlabeled data is added? 1 / 1 point

- Accuracy becomes unpredictable
- No change in accuracy
- Increase in accuracy
- Decrease in accuracy (selected)

Nice work

If the assumptions about the underlying class structure are incorrect, adding unlabeled data can lead to a classifier that misrepresents the true distribution of the classes, resulting in decreased accuracy.

2. What is the role of the kernel function in the label propagation algorithm? 1 / 1 point

- Calculates the final label assignment
- Optimizes the computational efficiency
- Determines the number of labels
- Controls label influence propagation (selected)

Nice work

The kernel function in label propagation, such as the radial basis function, determines how the influence of labels is propagated through the graph, affecting the smoothness and reach of label spread.

3. In a graph-based classification method, what would likely happen if two nodes are close together in the graph? 1 / 1 point

- They would have similar labels (selected)
- They would have different labels
- They would not influence each other
- They would be unrelated in the graph

Nice work

According to the assumption of graph-based methods, labels vary smoothly over the graph, meaning that nodes that are close together tend to have similar labels.

4. What is the primary role of the regularizer in label propagation? 1 / 1 point

- Increases the speed of convergence
- Maximizes the accuracy of labels
- Enforces smoothness of labels over the graph (selected)
- Reduces computational complexity

Nice work

The regularizer in label propagation ensures that the labels change smoothly over the graph, which helps in maintaining consistency and reliability in the label predictions across similar nodes.

5. How does the 'clamping' feature in label propagation influence the prediction outcomes? 1 / 1 point

- Enhances prediction accuracy
- Increases label diversity
- Keeps original labels fixed (selected)
- Reduces computational complexity

Nice work

In label propagation, the clamping feature ensures that the input labels remain unchanged in the output predictions, thereby maintaining the integrity of the original labeled data.

6. What is the main assumption in using graph-based methods for semi-supervised learning in sentiment categorization? 1 / 1 point

- Reviews should be independent of each other
- Similar reviews should have similar ratings (selected)
- All reviews require manual labeling
- Ratings are irrelevant to the text content

Nice work

The assumption in graph-based semi-supervised learning for sentiment categorization is that reviews with similar content (words used) are likely to have similar ratings, which guides the label propagation algorithm.

7. In the context of graph-based semi-supervised learning using a k-nearest neighbors (K-NN) kernel, what would be the impact of using a higher value of the parameter k in the kernel? 1 / 1 point

- Sparse matrix with less detailed connections
- Uniform node degree across the graph
- Increased computational complexity with no change in connections
- A more dense adjacency matrix with more extended connections (selected)

Nice work

The graph becomes more connected, as each node is connected to a larger number of neighbors. As a result, even distant nodes might be connected if they share a large number of common neighbors.

8. What is the effect of having a disconnected subgraph when using label propagation? 1 / 1 point

- Label propagation is slower
- Label propagation requires more iterations
- Label propagation does not work (selected)
- Label propagation becomes more accurate

Nice work

If there is a disconnected subgraph, label propagation cannot infer labels for its nodes because there is no path connecting these nodes to any labeled node in other parts of the graph.